

A Multi-Model Dataset for BOSCH Plasma-Etching: Optical Emission Spectra, Process Parameters, and Wafer Measurements for Data-Driven Plasma Modeling

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Introduction

The availability of high-quality process datasets in the public domain is crucial for advancing semiconductor manufacturing research. Plasma-based etching processes, such as the BOSCH process, are highly sensitive to prior chamber states and conditioning effects, which can introduce drifts and variations in wafer results. Despite this importance, comprehensive open datasets capturing both in-situ plasma behavior and ex-situ wafer characterization are still rare.

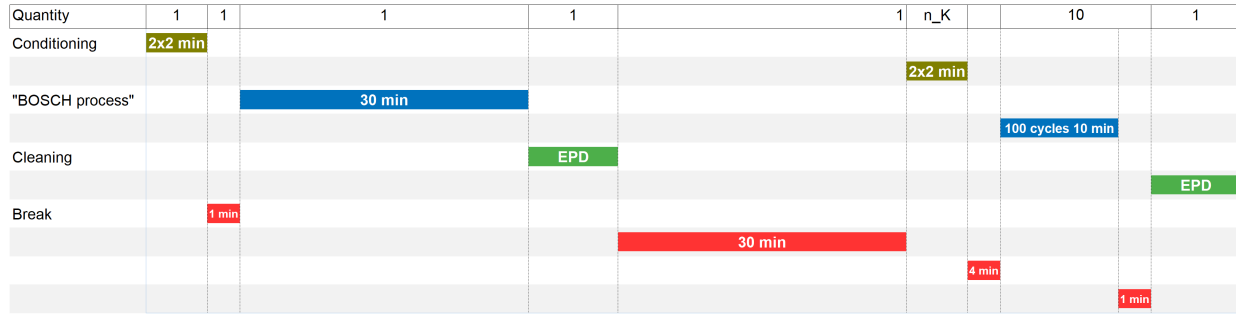
To address this gap, experiments were conducted at the Center for Microtechnologies (ZFM), Chemnitz, Germany, with a systematic Design of Experiments (DOE) focusing on the role of conditioning. A two-stage cleaning step (O_2 plasma followed by O_2/SF_6 plasma) was repeated under controlled variations: once, three times, or nine times, with conditioning performed either directly on the tool chuck, on a blank Si wafer, or on a black SiO_2 wafer. Each conditioning sequence was followed by processing a lot of 10 wafers without intermediate cleaning, thereby deliberately inducing different chamber states and process drifts.

While no clear trend could be established on the conditioning influence itself, the dataset is uniquely rich in information. It provides synchronized Optical Emission Spectroscopy (OES), process parameters, and detailed wafer measurements, enabling deeper investigations into plasma cycle structures and their relation to etch performance. Researchers can leverage this dataset to develop models for predicting etch depth, post-etch wafer uniformity, and selectivity with respect to the mask, as well as for advancing machine learning methods in etch process control.

1. DESIGN OF EXPERIMENT

The experiments were conducted using the Omega i2L DSi Rapier system manufactured by SPTS. The research focused on a BOSCH process that utilizes SF_6 as the etching gas and C_4F_8 as the passivation gas. The process begins with a 1-second ignition step, followed by 100 cycles comprising 4.5 seconds of the SF_6 phase and 1.5 seconds of the C_4F_8 phase. Each processing lot consists of 10 silicon wafers with a diameter of 200 mm and a $\langle 100 \rangle$ crystallographic orientation. The wafer design features over 99.5% exposed silicon area, organized into 89 squares, each measuring 1 mm^2 , along with a $1\text{ }\mu\text{m}$ thick layer of SiO_2 mask (please refer to ‘Wafer_layout.pdf’).

To prepare the system, a two-stage cleaning process is implemented. Figure 1 shows the detailed process flow for every wafer. The first step involves 2 minutes of O_2 plasma, followed by a second step with a combined O_2 and SF_6 plasma for another 2 minutes. After the cleaning, there is a 1-minute break, followed by a 30-minute BOSCH process serving as a dummy run. This is then succeeded by a final cleaning step that utilizes a combined O_2 and SF_6 plasma, controlled by endpoint detection. After a 30-minute break, conditioning steps are performed, mirroring the initial two-stage cleaning process. This conditioning is repeated once, three times, and nine times, with a 4-minute break between each repetition, done on the system chuck, blank Si wafer, or a blank SiO_2 wafer. Finally, the 10 wafers are processed sequentially, with a 1-minute break between each wafer. Please refer to ‘Lot_status.xlsx’ to link each lot with the experiment date and the conditioning steps. ‘C’, ‘Si’, and ‘ SiO_2 ’ refer to Chuck, Si wafer, and SiO_2 wafer, respectively. Due to some failure of the etching tool, the experiments with SiO_2 wafer and 3 conditioning were repeated. Also, the last 4 wafers of the lot 7 were damaged, so data for these wafers is not available.



EPD ... Cleaning process with endpoint detection

Figure 1. Process flow for every lot

2. DATA DESCRIPTION

2.1 Optical Emission Spectroscopy (OES)

The emission measurement for the bulk of the plasma was carried out using Optical Emission Spectroscopy (OES HR4-4UVV250-5 by OceanOptics), which provides 3648 spectral channels across a wavelength range of 185.89 nm to 883.97 nm. All intensity measurements were continuously recorded and stored in a database; however, some irregularities and missing samples were present, so the exact timestamp of each measurement is included. The original double-precision data was compressed into a 16-bit lossless format using dictionary encoding, with a single dictionary file, 'Dictionary_OES.nc', applied consistently across all days for intensity values. Each daily '.nc' file contains NetCDF groups corresponding to the wafers processed on that day, with individual groups for each wafer containing emission wavelengths, timestamps, and intensity data. A Python code snippet is provided below to demonstrate how to efficiently access the OES data files. All in-situ data is stored in NetCDF (.nc) format, ensuring compatibility across multiple programming platforms.

```
import pandas as pd
from netCDF4 import Dataset, num2date

with Dataset("Dictionary_OES.nc", "r") as dict_ds:
    decoder = dict_ds["data"][:]

with Dataset("Day_2024_07_02.nc", "r") as ds:
    # Show available groups
    print(ds.groups)

    # Get group for wafer 2
    grp = ds.groups['Wafer_02']

    #Read and decode data into arrays
    encoded = grp["data"][:]
    data = decoder[encoded]
    times = num2date(grp["times"][:], grp["times"].units, grp["times"].calendar,
                    only_use_cftime_datetimes=False,
                    only_use_python_datetimes=True)
    cols = grp["wavelengths"][:]

# Turn arrays into a dataframe
df = pd.DataFrame(data, index=times, columns=cols)
```

2.2 Process Parameters

Process parameters such as gas flow, pressure, temperature, and power are monitored using specialized sensors, including mass flow controllers, pressure transducers, thermocouples, and power meters. These parameters are continuously recorded by Data Acquisition (DAQ) systems at a sampling rate of 5 Hz. Unlike the OES data, the process parameter data contains no irregular or missing samples and is not divided by days; instead, the complete dataset is stored in a single file, 'Process_data.nc'. Similar to the OES data, it is encoded in a 16-bit lossless format using dictionary encoding, with a dedicated dictionary file, 'Dictionary_process.nc', for all wafers. Each wafer has its own group within the file, which contains the parameter names, corresponding timestamps, and sensor values. A sample Python code snippet is provided below to demonstrate how to access and load the process parameter data.

```
import pandas as pd
from netCDF4 import Dataset, num2date

with Dataset("Dictionary_process.nc", "r") as dict_ds:
    decoder = dict_ds["data"][:]

with Dataset("Process_data.nc", "r") as ds:
    # Show available groups
    print(ds.groups)

    # Get group for wafer 2
    grp = ds.groups['Day_2024_07_02_Wafer_02']

    #Read and decode data into arrays
    encoded = grp["data"][:]
    data = decoder[encoded]
    times = num2date(grp["times"][:], grp["times"].units, grp["times"].calendar,
                    only_use_cftime_datetimes=False,
                    only_use_python_datetimes=True)
    cols = grp["feature"][:]

# Turn arrays into a dataframe
df = pd.DataFrame(data, index=times, columns=cols)
```

2.3 Measurement data

The measurement data were collected twice: first on the day of the experiment ('Lot_status.xlsx'), when 9 points were measured, and second during February 2025, when all 89 points were measured.

2.3.1 Measurement of 9 Points

Before and after processing, the thickness of the wafer and the SiO₂ mask was measured at nine predefined locations ('Wafer_layout.pdf') using two instruments: a reflectometer (MPROBE™ 40-MSP by Semiconsoft) and a profilometer (Dektak® 8 Stylus Profiler by Veeco). The reflectometer was used to measure the SiO₂ mask thickness, while the profilometer determined the step height between the SiO₂ mask and the underlying silicon wafer surface. By comparing the pre- and post-process measurements, the etch depth of both Si and SiO₂ was calculated at each location. The file 'Si_Oxide_etch_9_points.csv' contains the raw measurements, the calculated etch depths, and the corresponding measurement points; all values are in µm. Since these measurements were performed manually by an operator for each location on every wafer, a certain degree of human error is expected.

2.3.2 Measurement of 89 Points

Similar to 9 points, the 89-point measurements were taken with two instruments: an interferometer (FTR MicroProf® 300, FRT) for post-etch oxide thickness and a profilometer (Tencor® P-17, KLA) for step height. Pre-etch thickness was measured similarly to 9 points using a reflectometer, but only 15 data points were available (Figure 2). The missing points were interpolated via Inverse Distance Weighting¹ (IDW, $p=2$). This method weights nearby points more strongly than distant ones to maintain a spatial data structure. For the pre-etch oxide thickness measurement, the etch variation across all wafer is very small, 991.6 - 1024.1 nm, any averaging technique should work fine.

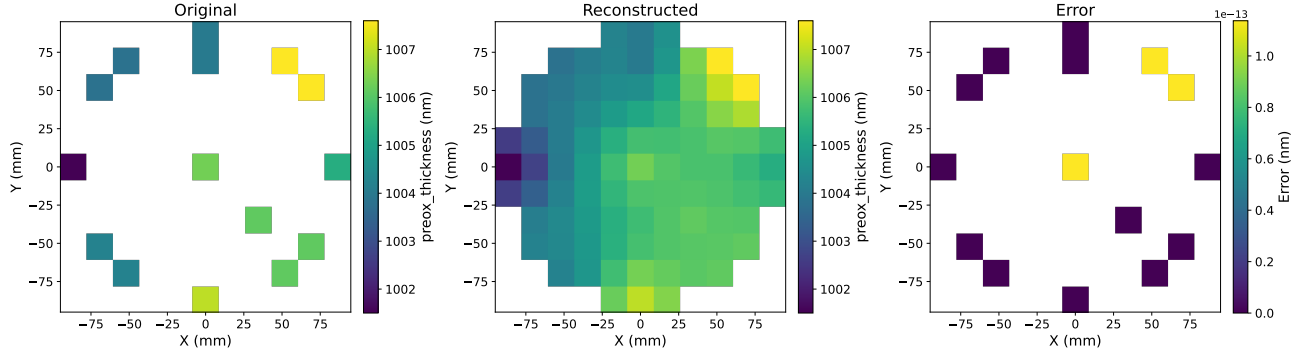


Figure 2. IDW interpolation for pre-etch oxide thickness for wafer number ‘2024-07-09_06’

Post-etch thickness from the interferometer is based on spectral interference of reflected light. If the fit fails, the system returns NaN values, which were again interpolated using IDW. Figure 3 shows an example wafer (2024-07-09_06’) with maximum NaNs.

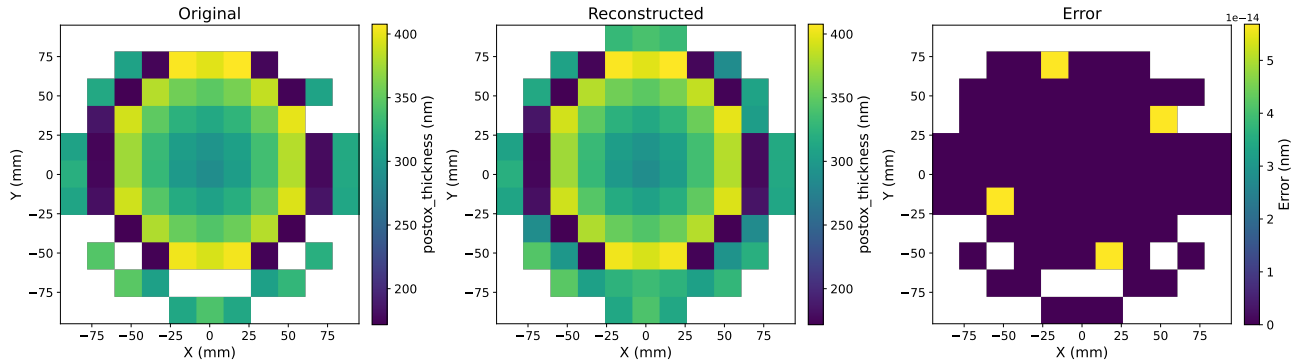


Figure 3. IDW interpolation for post-etch oxide thickness for wafer number ‘2024-07-09_06’

Raw measurements and the calculated etch thickness are provided in ‘Si_Oxide_etch_9_points.csv’ (9 points), ‘Si_Oxide_etch_89_points.csv’ (89 points), and missing data information is available in ‘Lot_status.xlsx’.

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REFERENCES

- [1] Panigrahi, N., [Inverse Distance Weight], 1–7, Springer International Publishing, Cham (2020).